

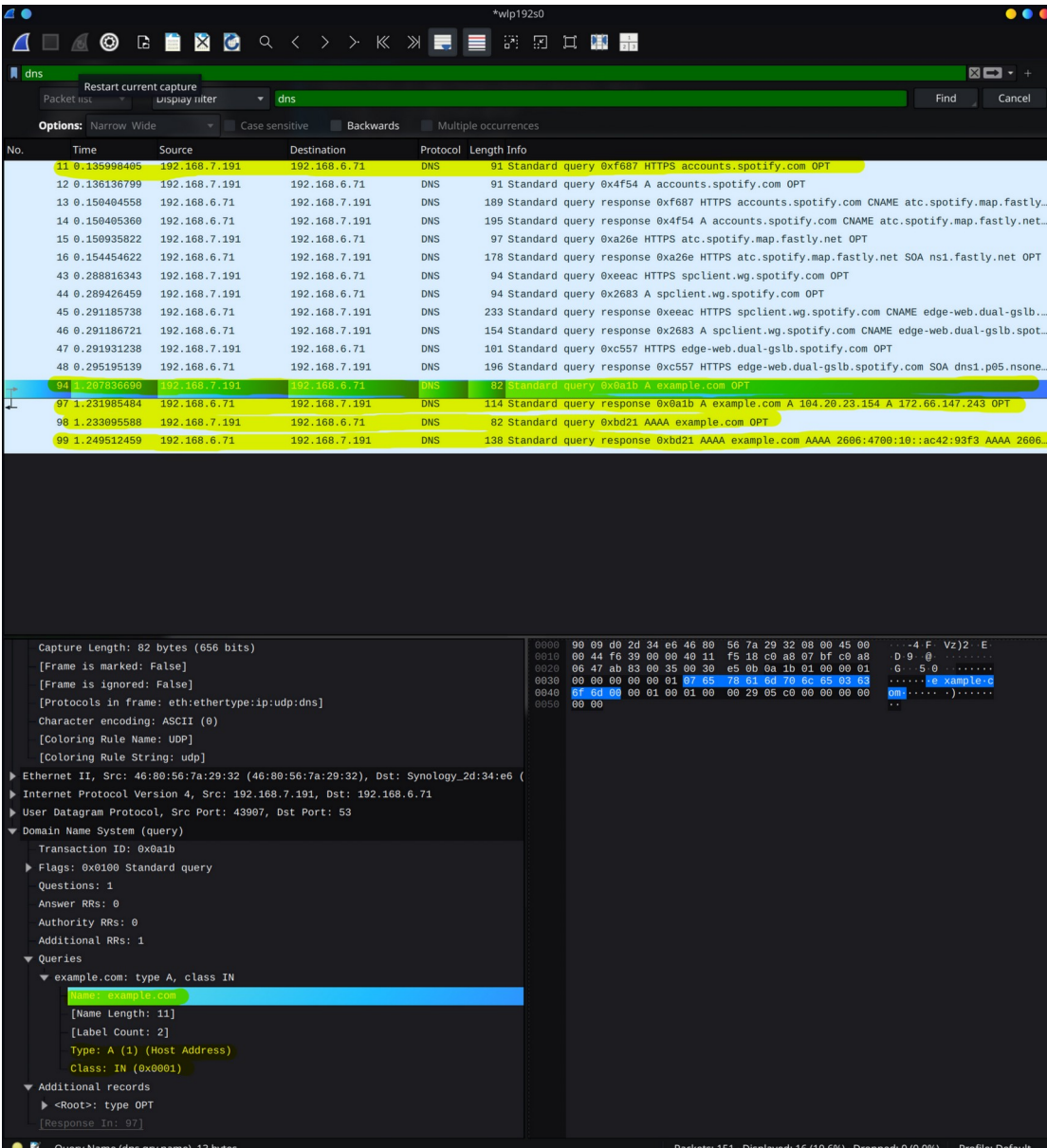
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CS-158A

Lab 2

Checkpoint 1:

4 DNS packets appeared with the target IP of 192.168.6.71, which is the IP of the local DNS server on my home network. Other DNS packets were sent to Spotify’s servers from the client running on my laptop. (Pictured: 4 queries; resolver ip 192.168.6.71; name example.com; type A; class IN)



Checkpoint 2:

I recorded a TTL of 2 minutes 56 seconds. When that timer expires, the record does as well and I have to fetch a fresh one from DNS. Pictured below are Name: example.com; Type A; Class IN; TTL 2 min 56 sec.

46	0.291186721	192.168.6.71	192.168.7.191	DNS	154 Standard query response 0x2683 A spclient.wg.spotify.com CNAME edge-web.dual-gslb.spotify.com OPT
47	0.291931238	192.168.7.191	192.168.6.71	DNS	101 Standard query 0xc557 HTTPS edge-web.dual-gslb.spotify.com OPT
48	0.295195139	192.168.6.71	192.168.7.191	DNS	196 Standard query response 0xc557 HTTPS edge-web.dual-gslb.spotify.com SOA dns1.p05.nsona.com OPT
94	1.207836690	192.168.7.191	192.168.6.71	DNS	82 Standard query 0xa01b A example.com OPT
97	1.231985484	192.168.6.71	192.168.7.191	DNS	114 Standard query response 0xa01b A example.com A 104.20.23.154 A 172.66.147.243 OPT
98	1.233095588	192.168.7.191	192.168.6.71	DNS	82 Standard query 0xbd21 AAAA example.com OPT
99	1.249512459	192.168.6.71	192.168.7.191	DNS	138 Standard query response 0xbd21 AAAA example.com AAAA 2606:4700:10::ac42:93f3 AAAA 2606:4700:10::ac42:93f3

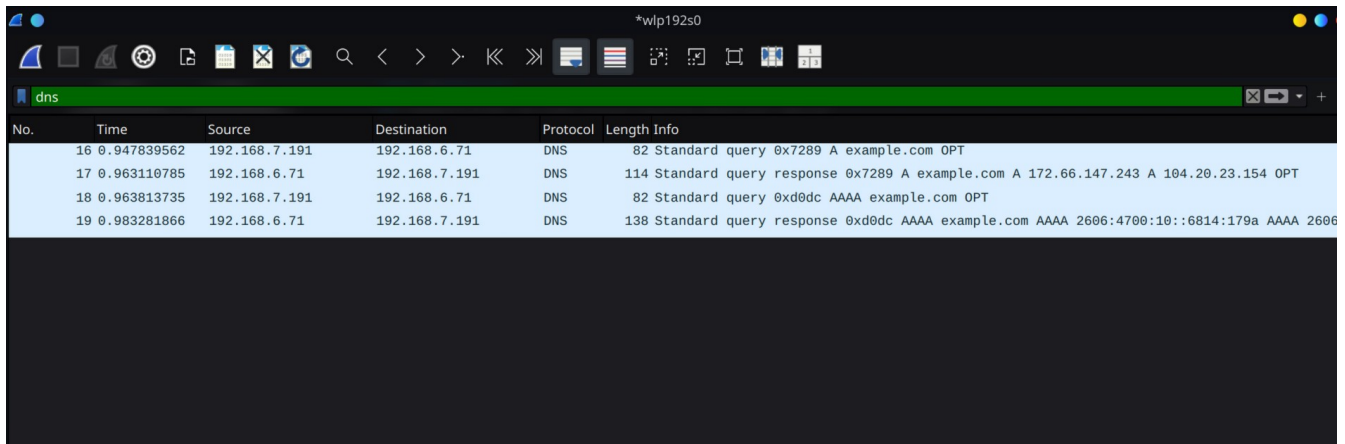
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Ethernet II, Src: Synology_2d:34:e6 (90:09:d0:2d:34:e6), Dst: 46:80:56:7a:29:32 (0000 46 80 56 7a 29 32 90 09 d0 2d 34 e6 08 00 45 00 F.Vz)2...-4...E
Internet Protocol Version 4, Src: 192.168.6.71, Dst: 192.168.7.191 (0010 00 64 f8 65 40 00 40 11 b2 cc c0 a8 06 47 c0 a8 d e@ @ .....G...
User Datagram Protocol, Src Port: 53, Dst Port: 43997 (0020 07 bf 00 35 ab 83 00 50 fa f7 0a 1b 81 80 00 01 ...5...P.....
Domain Name System (response) (0030 00 02 00 00 00 01 07 65 78 61 6d 70 6c 65 03 63 .....e xample.c
Transaction ID: 0x0a1b (0040 6f 6d 00 00 01 00 01 c0 0c 00 01 00 01 00 00 00 om...h.....
Flags: 0x8180 Standard query response, No error (0050 b0 00 04 68 14 17 9a c0 0c 00 01 00 01 00 00 00 ...h.....
Questions: 1 (0060 b0 00 04 ac 42 93 f3 00 00 29 04 d0 00 00 00 00 ...B.....).....
Answer RRs: 2 (0070 00 00 .....
Authority RRs: 0
Additional RRs: 1
Queries
  example.com: type A, class IN
    Name: example.com
    [Name Length: 11]
    [Label Count: 2]
    Type: A (1) (Host Address)
    Class: IN (0x0001)
Answers
  example.com: type A, class IN, addr 104.20.23.154
    Name: example.com
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 176 (2 minutes, 56 seconds)
    Data length: 4
    Address: 104.20.23.154
  example.com: type A, class IN, addr 172.66.147.243
Additional records

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Checkpoint 3:

I predicted that my OS would need a fresh query rather than using the cached data because it has been longer than the TTL on the response packet. This matched what I saw running the query again.



No.	Time	Source	Destination	Protocol	Length	Info
16	0.947839562	192.168.7.191	192.168.6.71	DNS	82	Standard query 0x7289 A example.com OPT
17	0.963110785	192.168.6.71	192.168.7.191	DNS	114	Standard query response 0x7289 A example.com A 172.66.147.243 A 104.20.23.154 OPT
18	0.963813735	192.168.7.191	192.168.6.71	DNS	82	Standard query 0xd0dc AAAA example.com OPT
19	0.983281866	192.168.6.71	192.168.7.191	DNS	138	Standard query response 0xd0dc AAAA example.com AAAA 2606:4700:10::6814:179a AAAA 2606

Checkpoint 4

- I do not see the queries directly. It is routed through my local DNS server and returns a pointer to dns.google
- The RD and RA values were both 1. This combined with just the 2 calls

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▼ Flags: 0x8180 Standard query response, No error
  1... .. = Response: Message is a response
  .000 0... .. = Opcode: Standard query (0)
  .... .0.. .. = Authoritative: Server is not an authority for domain
  .... ..0. .... = Truncated: Message is not truncated
  .... ...1 .... = Recursion desired: Do query recursively
  .... .... 1... .. = Recursion available: Server can do recursive queries
  .... .... .0.. .. = Z: reserved (0)
  .... .... ..0. .... = Answer authenticated: Answer/authority portion was not
  .... .... ...0 .... = Non-authenticated data: Unacceptable
  .... .... .... 0000 = Reply code: No error (0)
```

RD and RA are both 1